PARTH SANTOSH CHITTALWAR

Computer Technology Engineering Undergraduate
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EDUCATION

Priyadarshini College of Engineering — B.Tech, Computer Technology Expected June 2028
CGPA: 9.1 / 10.0 | SSC: 83.60% | HSC: 75.50%

TECHNICAL SKILLS

Languages: C, C++, JavaScript, TypeScript
Frontend: HTML5, CSS3, React, Tailwind CSS, React Router, Responsive Design
Backend: Node.js, Express.js, REST APIs, JWT Authentication
Database: MongoDB, SQL, Mongoose
Core CS: Data Structures, Algorithms, Object-Oriented Programming, DBMS, Operating Systems
Developer Tools: Git, GitHub, VS Code, Docker, Vite, npm, Render
AI & Automation: Google Gemini AI, Puppeteer

PROJECTS

AI Interview Preparation Platform github.com/ParthChittalwar/Interview-Intelligence • interview-intelligence-8l7z.onrender.com
React · Vite · Tailwind CSS · Node.js · Express.js · MongoDB Atlas · Mongoose · JWT Authentication · Google Gemini AI · Puppeteer
– Built a full-stack AI Interview Preparation Platform powered by Google Gemini AI, generating technical questions, behavioral questions, skill-gap analysis, and personalized roadmaps.
– Implemented secure JWT authentication with HTTP-only cookies and protected routes, backed by MongoDB Atlas and Mongoose for persistent user data.
– Built resume upload and parsing (PDF/DOCX) with AI-driven analysis and automated interview-report generation via Puppeteer.
– Deployed frontend and backend independently on Render with a MongoDB Atlas cloud database for production use.

NovaOS – 64-bit Operating System github.com/ParthChittalwar/NovaOS-Operating-System
C++20 · NASM Assembly · UEFI · Limine Bootloader · x86_64 Architecture · CMake · QEMU · Framebuffer Graphics
– Developed NovaOS, a modular 64-bit operating system from scratch in C++20 and NASM Assembly for the x86_64 architecture.
– Implemented UEFI booting via the Limine protocol with a higher-half kernel, GDT/IDT setup, interrupt handling, a physical memory manager, and a kernel heap allocator.
– Built core OS components — framebuffer console, PS/2 keyboard driver, PIT timer, round-robin scheduler, and a virtual file system (VFS) — plus an interactive kernel shell with panic handling.
– Built and tested the OS using QEMU with a CMake-based, cross-platform build system.

SketchBoard-Pro github.com/ParthChittalwar/SketchBoard-Pro • sketchboard-pro.pages.dev
React · TypeScript · Vite · Tailwind CSS · Canvas API · TanStack Router · Framer Motion
– Architected an infinite, zoomable canvas rendering engine from scratch with the Canvas API, supporting smooth pan/zoom, undo/redo, and multi-tool drawing at scale.
– Built a modular, extensible architecture separating tools, shapes, and export pipelines (PNG/JPG/SVG/PDF) so new features ship without touching core rendering logic.
– Optimized canvas rendering and autosave performance to keep interactions responsive under heavy shape counts while supporting full offline use.
– Delivered a presentation mode with a laser pointer and coordinate-grid overlay, extending the app into a lightweight teaching tool.

Personal Portfolio Website github.com/ParthChittalwar/Portfolio • portfolio-5yf.pages.dev
React · TypeScript · Tailwind CSS · Framer Motion · React Three Fiber
– Architected an interactive 3D hero using React Three Fiber with code-split, lazy-loaded scenes, cutting initial load time while holding smooth 60fps interaction.
– Built a command palette and dynamic project-page system with client-side search and filtering for instant navigation across the project catalog.
– Engineered accessibility and SEO into the component structure (semantic markup, ARIA labeling, meta tags), keeping rendering consistent across browsers and devices.

Democratic Bharat V2.0 github.com/ParthChittalwar/Democratic-Bharat-Voting-Awareness-Platform
React · TypeScript · Tailwind CSS · React Router · Lucide React
– Developed an interactive EVM and VVPAT simulation engine modeling real election workflows into a step-by-step visual experience.
– Implemented multi-language support and a modular content architecture to deliver voting-awareness modules to a broader audience.
– Optimized component structure for responsive rendering across devices, maintaining a low-friction UX for first-time voters.

ACHIEVEMENTS

– Maintained a 9.1/10 CGPA in B.Tech Computer Technology.
– Built and deployed multiple production-grade full-stack and frontend applications featuring AI integration, cloud deployment, and modern web technologies.
– Regularly practice Data Structures & Algorithms on LeetCode and GeeksforGeeks.

CERTIFICATIONS

C++ Programming (Udemy) | Data Structures & Algorithms (GeeksforGeeks) | Web Development Bootcamp (The Odin Project) | Git & GitHub Essentials (LinkedIn Learning)